Resident OS

Decisions with receipts.

AI Operations Layer for Apartment Communities — System Architecture

From a resident's message to an audited, cited, human-approved decision

Google Cloud Run + Vercel

Supabase Postgres · pgvector · LangGraph

Users

Resident
Reports issues, tracks status, asks policy questions

Property Manager
Clears the approval queue, overrides, audits

Maintenance Tech
Receives job, parts list, access notes

Vendor
Accepts or declines via signed link — no account

Asset Owner
Monthly cost, SLA, AI governance report

ML Engineer (admin)
Ops console: agents, evals, failures, drift

ROLE MODEL

RBAC × scope (org / property / building / unit), enforced in an API dependency **and** Postgres RLS.

Frontend — Next.js 15

FIVE SURFACES

- **/app** resident, mobile-first
- **/manage** dense, keyboard-first
- **/owner** reporting, print-adjacent
- **/tech** large targets, offline-tolerant
- **/v/[token]** vendor, no auth
- **/ops** ML console, admin only

TRANSPORT

Server Components for reads. Route Handlers act as a BFF, so the browser never holds a service key. **SSE** streams step events — the timeline renders immediately with pending steps hollow.

SIGNATURE UI

- Status lamp — priority, never colour alone
- Evidence rail — citation chips, two-way hover
- **Blue** = machine decided · **Brass** = human authorised

Backend API — FastAPI

`POST /v1/tickets`	idempotency key required
`GET /tickets/{id}/stream`	SSE step + token events
`GET /v1/approvals`	queue, ordered by SLA risk
`POST /approvals/{id}/…`	approve · reject reassign
`GET /v1/runs/{id}`	full execution trace
`GET /runs/{id}/replay`	re-run against pinned versions
`GET /knowledge/search`	hybrid retrieval with scores
`GET /v1/ops/*`	health · agents · failures · drift
`POST /webhooks/vendor`	untrusted → quarantine

REQUEST PIPELINE

auth → tenancy (RLS context) → rate limit → **idempotency** → handler → RFC 7807 errors. Agentic systems retry; without idempotency keys a retry becomes a duplicate real-world action.

LangGraph Agent Workflow — the maintenance loop

1 **Ingest & acknowledge** `code`
Persist raw, return ticket number in under 1s. No model call on this path, ever.

2 **Input guardrails** `code`
PII redact · injection scan · toxicity · protected-attribute scrub

3 **Safety Sentinel** `rules` `haiku`
Rules
OR
model — never AND. Target recall 0.99, precision 0.70.

→ **P0 protocol** `code`
Page on-call: push + SMS + voice, simultaneous, quiet hours ignored.
No model between detection and the phone.

4 **Intake Normalizer** `haiku`
Messy text + photo captions → TicketFacts. Strict schema, one repair retry.

5 **Context Broker** `code`
Parallel retrieve: policy · unit history · asset · memory. Rank, compress, budget to 8k with provenance IDs.
See page 2.

6 **Triage router** `code`
Deterministic blast-radius score → solo or council. Target council rate 8–12%.

7 **Diagnostician** `sonnet`
Grounded cause + actions.
Every claim carries a citation ID
or moves to unknowns.

7b **Council — 3 members in parallel** `sonnet ×3`
Split by
evidence, not persona
: policy · history · cost. Deterministic reviewer, then synthesis with dissent recorded.

8 **Dispatch Planner** `sonnet`
Trade, vendor, window, parts, cost.
Proposes only. Never writes.

9 **Policy Auditor** `rules` `haiku`
The adversary. Sees the decision and the policy — deliberately not the reasoning chain. statute > lease > SOP > contract.

10 **Decision gate** `code`
Calibrated confidence → auto (≥0.85) · approval (0.60–0.85) ·
escalate (<0.60)

11 **Human approval — LangGraph interrupt()** `human`
The approval queue *is* an interrupted graph. Single-use, action-scoped token released on approve. Every reject is a labelled training example.

12 **Communicator** `haiku`
Template-bound copy. Never receives protected attributes — it cannot leak what it never saw.

13 **Output guardrails** `code`
Fair housing / ADA · PII leak · citations resolve · numeric sanity. A block routes to a human,
never a silent rewrite
.

14 **Execute + audit** `code`
Work order, vendor dispatch, resident message — all idempotent. Immutable decision record written.

15 **Judge** `async`
Offline 100% in CI, online 10% sample. Different model family than generation, version pinned.

Design constraints

- ✓ **≤ 5 model calls** on the critical path. 3 solo, 5 with council.
- ✓ **Reliability compounds**: 10 steps at 95% = 60% end to end.
- ✓ **Read parallel, write single-threaded.** Only the orchestrator holds write tools.
- ✓ **Model proposes, code disposes.** Every consequential outcome passes through testable code.
- ✓ **Calibrated abstention** is a first-class output.

Agent roster

Component	Tier	Writes
Orchestrator	`code`	**all**
Safety Sentinel	`rulesvsm`	no
Intake Normalizer	`small`	no
Context Broker	`code`	no
Diagnostician	`mid`	no
Policy Auditor	`rules+sm`	no
Dispatch Planner	`mid`	proposes
Communicator	`small`	no
Judge	`mid`	no

Split by **blast radius, context need, and evaluability** — never by job title. Permissions enforced server-side in a grant map, not in a prompt.

Priority, SLA & outcome

P0 life safety	`page immediately`
P1 habitability	`24h statutory`
P2 urgent	`3 days`
P3 routine	`7 days`

SLA derived from category × **property jurisdiction**. Habitability windows are state law, not preference.

OUTCOME

- ✓ Fewer wrong truck rolls (~$150 each)
- ✓ Faster P0 — habitability exposure
- ✓ An audit record that did not exist
- ✓ ~$0.023/ticket · ~$73/property/year

3 model calls typical · **5 max on critical path** · **human approves** · **every claim cited**

The AI Internals — Retrieval, Context, Memory, Guardrails

How evidence gets into the model, and how a claim becomes a citation

Knowledge base — as code

knowledge//*.md in git.** A PR that edits a policy re-embeds only changed files and **runs the eval suite**. A policy change that breaks retrieval fails CI.

REQUIRED FRONT MATTER

```
id: SOP-PLM-04
version: 3
effective_from: 2025-01-01
effective_to: null
jurisdiction: [US-VA, US-MD]
authority: internal_sop
supersedes: SOP-PLM-03
```

CONFLICT PRECEDENCE — IN CODE, NOT IN A PROMPT

`statute > lease > internal_sop > vendor_contract`

CHUNKING, PER CORPUS

Leases	clause-level, keep section path
SOPs	heading-aware, 300–600 tok, 15% overlap
Rules	one rule per chunk
History	whole ticket; embed symptom + resolution

Retrieval pipeline — every stage earns its place

Query expand	BM25 / tsvector	Dense / pgvector	RRF fusion	Metadata filter	Cross-encoder	Parent expand	Compress
acronym + synonym map. **No model call**	GIN index, top 30. Embeddings smear identifiers — **SOP-PLM-04** and **§7.3** are what people cite	HNSW m=16, halfvec, top 30. Catches paraphrase	1/(60+rank). Rank-based, so no score normalisation between two incomparable systems	**After** fusion. Pre-filtering a selective predicate collapses HNSW recall	bge-reranker-v2-m3, 30→6. Query and doc attend **together**	chunk → containing section. A clause is meaningless without its conditions	extractive sentence selection. **Never abstractive** on policy text

BI-ENCODER VS CROSS-ENCODER — WHY BOTH

A **bi-encoder** embeds query and document separately, so the document was encoded without knowing the query. Fast enough to scan 200k chunks, imprecise. A **cross-encoder** takes (query, document) as one input and attends across both — precise, but one model run per candidate.

So: bi-encoder for **recall** over the corpus, cross-encoder for **precision** over 30. The reranker stays loaded in-process; per-request loading would dominate the latency budget.

DELIBERATELY NOT USED

- × **Knowledge-graph RAG** — our graph is relational with clean foreign keys. A SQL join is exact and free.
- × **Dedicated vector DB** — under ~10M vectors Postgres is faster end to end and one less system. We have ~200k chunks.
- × **HyDE** — one extra model call for modest gain in a small controlled vocabulary.

MEASURED BEFORE ANY AGENT EXISTED (PHASE 2)

Strategy	recall@5	nDCG@10
Dense only	~0.62	–
Lexical only	~0.55	–
Hybrid RRF	**~0.78**	–
+ cross-encoder	**~0.84**	–

Retrieval is measured before agents are built. Retrieval quality caps everything downstream — if recall@5 is 0.6 you will spend weeks blaming prompts for a retrieval problem.

Rerank lift is measured and published. **If it is not positive, reranking is deleted.** Same rule as every other component.

Context engineering — the provenance envelope

The highest-leverage anti-hallucination technique here, and it costs almost nothing.

```
[C1] (SOP-PLM-04 · v3 · effective 2025-01-01 · §2)
     After two failed drain-clearing attempts within
     12 months, replace the P-trap assembly rather
     than repeating mechanical clearing.

[C2] (lease · unit 4B · §7.3)
     Owner is responsible for repairs arising from
     normal wear and tear...

[F1] (fact: asset) unit=4B · fixture=kitchen_sink
     installed=2019-03 · warranty_expires=2027-03
```

Instruction: every factual claim must carry a bracketed source ID. If no source supports it, put it in unknowns instead.

Then deterministic code verifies — not a judge: every cited ID exists · every sentence with a number, date, dollar or § carries one · cited figures appear verbatim in the referenced chunk. One targeted regeneration, then a human.

8,000-TOKEN BUDGET, PER SLOT

system + schema `cached`	1,000
policy / SOP excerpts	1,600
unit + asset facts `key-value`	700
similar resolved cases (k=3)	1,300
conversation summary	500
current ticket `never truncated`	600
output reserve	**2,300**

Filtered by the **ticket's timestamp**, not now(). Replaying a March decision uses March's policy — this is what makes the audit trail legally meaningful.

Memory — four stores, one interface

Store	Backing	Write policy
Working	LangGraph checkpoint	automatic
Entity	Postgres tables	authoritative only
Episodic	pgvector	on ticket close
Semantic	KB in git	via PR
Reflection	Postgres + pgvector	**human approves**

MEMORY WRITES ARE PROPOSALS

An agent calls `memory.propose(claim, evidence, scope, confidence)`. Nothing enters durable memory without human approval or N-fold independent confirmation.

Why: an agent that writes freely to its own memory will eventually write something wrong, retrieve it as fact, and reinforce it. Memory poisoning is the scariest failure mode in long-running agent systems.

RANKING

```
relevance × recency_decay(180d)
 × scope_weight(unit 1.0 > bldg 0.8 > prop 0.6)
 × confirmation_count^0.3
 × (0 if contradicted_by_newer)
```

FACTS SUPERSEDE, NEVER OVERWRITE

A conflicting fact sets `valid_to` on the old row and inserts a new one. History is preserved — required for replay and audit. Two timestamp columns give the temporal validity a graph memory product sells, because our entities are already a schema.

EVALUATED, NOT ADOPTED

Mem0 / Zep / LangMem solve fuzzy recall across open-ended conversation. Our durable facts are structured and legally consequential — a warranty date must be exactly right, and a summariser that mangles it is a defect. A `MemoryProvider` interface keeps an adapter one flag away.

Guardrails — where each executes

Stage	Guardrail
`API edge`	auth / tenant scope
`API edge`	rate + size limits
`pre-model`	**PII detect & redact**
`pre-model`	**injection scan**
`pre-model`	toxicity / abuse
`pre-model`	**protected-attribute scrub**
`post-model`	schema validation
`post-model`	**citation verification**
`post-model`	numeric sanity
`post-decision`	**policy compliance**
`pre-send`	**fair housing / ADA**
`pre-send`	output PII leak
`pre-tool`	grant map + approval token
`gateway`	cost circuit breaker
`pre-write`	idempotency

INJECTION — ARCHITECTURE FIRST

Everyone scans the chat box and forgets that **vendor SMS replies, uploaded PDFs, OCR output and image captions enter the same context window**. The real control: untrusted content never reaches an agent holding write tools. The scanner is the second line.

FAIR HOUSING IS A CI JOB

Not a sentence in a system prompt. **Paired probes** identical except one protected-class signal — voucher, wheelchair, service animal, children, non-English name — asserting response **parity**. Hard fail on any violation. This is exactly the test a fair-housing testing organisation would run against you.

No protected-attribute column exists in the schema. Not nullable, not unused — the columns do not

Resident OS

Decisions with receipts.

Reliability, Evaluation, Cost & Deployment

How the system is measured, how it fails safely, and what it costs to run

~$0.023 / ticket

~$73 / property / year

Data layer — Postgres + pgvector

DOMAIN

```
orgs → properties → buildings → units
people · roles · tenancies · assets
tickets · ticket_media · work_orders
vendors · vendor_scores
```

THE AI SPINE — **THIS IS THE PRODUCT**

```
agent_runs → agent_steps → llm_calls
retrievals · guardrail_events
decisions → approvals
```

KNOWLEDGE & MEMORY

```
kb_documents → kb_chunks (halfvec + tsvector)
episodic_memory · reflections
```

EVALUATION & ML OPS

```
eval_datasets → eval_items · eval_runs
failure_events · prompt_versions
metric_rollups · label_queue
```

THREE RULES THAT MATTER

- **org_id on every row, RLS on every table.** A single unscoped query is a breach notification.
- **Append-only** on decisions, llm_calls, guardrail_events, with a nightly hash chain. Converts "we log things" into "we can prove what we logged".
- **Dashboards read metric_rollups**, never raw events. Observability that gets heavier as things degrade is worse than none.

Model gateway — policy vs transport

Buy the transport, build the policy. Building your own gateway is six weeks of proxy code that is not your differentiator.

POLICY LAYER — YOUR CODE, ~400 LINES

model routing by task class · context budget · retrieval strategy · cache decision · per-org cost budget + circuit breaker · prompt version pinning · A/B assignment

TRANSPORT — LITELLM SELF-HOSTED, COMMODITY

provider abstraction · fallback · retries · virtual keys · spend logging · OTEL export

Tier	Model	Used for
small	haiku	safety, intake, comms, audit
mid	sonnet	diagnose, dispatch, council, judge
large	opus	escalated review, <1%
local	ollama	offline dev, no-vendor mode

COST LEVERS, IN IMPACT ORDER

- Prompt caching — ~90% off cached input, and it cuts **time-to-first-token** too
- Model routing — 5× spread between tiers
- Batch API — 50% off non-interactive work
- Output discipline — output costs 5× input
- Tool output compression — projection, ranked truncation **with a marker**, folding

Never semantic-cache a decision. Two similar tickets in different units are different decisions — that dispatches a plumber to the wrong apartment.

Failure & graceful degradation

Failure	Recovery
Schema violation	1 repair → human
Uncited claim	1 regen → human
Retrieval miss	broaden, retry, then "policy not found" + escalate
Tool timeout	2 retries, backoff + jitter
Provider 5xx	fallback provider, mark `model_substituted`, **force human approval**
Council member dies	synthesise from the rest, confidence −0.15
Loop / oscillation	hard stop at 12 nodes
Cost breach	downgrade tier → rules-only

THE DEGRADATION LADDER

full → **no council** → **no model diagnosis** → **rules only** → **queue + ack**

Rules-only mode is ~150 lines — keyword safety screen, category default SLA, fixed acknowledgment. With zero model providers reachable, a resident still gets a ticket number and a gas leak still pages the on-call. **The building keeps running when a vendor's API does not.**

LATENCY BUDGET, P50

acknowledgment	< 200 ms
input guardrails	< 120 ms
retrieval	< 350 ms
safety + intake	< 500 ms
diagnosis streamed	< 2,500 ms
total	**p50 <6s · p95 <20s**

Evaluation — four layers

Layer	Tool	Gates?
Unit — agents, guardrails as pure functions	pytest	**yes**
Component — recall@k, MRR, nDCG	ragas	**yes**
End-to-end — labelled golden set	deepeval	**yes**
Adversarial — fair housing, injection, PII	promptfoo	**hard fail**

THE GOLDEN SET — SKEWED TOWARD HARD CASES

Uniform sampling wastes labelling effort on tickets any system handles. Ambiguous priority · chargeback-disputable · life-safety **near-misses** ("smells like garbage" vs "smells like gas") · recurring · multilingual · missing-information where the correct answer is **to ask**.

Labelled by a human, before any prompt is tuned. Labelling after tuning encodes the model's behaviour as ground truth.

GATE WITH TOLERANCE BANDS, NOT THRESHOLDS

```
p0_recall     {min: 0.98, tol: 0.00}
priority_f1   {min: 0.82, tol: 0.02}
groundedness  {min: 0.93, tol: 0.02}
fair_housing  {max_violations: 0}
cost_per_ticket {max: $0.05}
```

Model nondeterminism makes exact gates flaky. **A flaky gate gets ignored, and an ignored gate is worse than no gate.**

JUDGE CAVEATS

- Judge with a **different model family** than you generate with — self-preference bias is real
- **Pin the judge version.** A silent upgrade invalidates your whole time series
- Human agreement sits ~85–92%. The judge is a **regression detector, not ground truth**

ML Ops console — /ops, admin only

One shell, one global time range. Debugging reads across agent health, cost, evals and gateway **simultaneously** — four separate routes hide the correlation that is usually the answer.

A Health verdict + 4 quadrants

quality · reliability · cost · human override rate. One line, computed, before anything else

B Agent scorecard

ranked by **degradation, not alphabetically**. Change narrative is rule-generated — a model writing it would invent causes

C Failure feed

typed events with why + fix. Promote → `label_queue`, **never straight into a dataset**

D Model + prompt registry

what is live, traffic split, per-version quality, one-click rollback by config flip

E Eval history + calibration

metric sparklines with band corridors; reliability diagram overlaying online vs offline

F Drift + output inspector

alert on online-vs-offline delta only; display the rest. Sample real outputs by filter

THE LOOP THAT MAKES IT IMPROVE

production failure → **human labels it** → **golden set** → **CI gate** → **regression prevented**

Deployment & observability

GOOGLE CLOUD RUN + VERCEL

Service	Config
web (Next.js)	`Vercel edge CDN`
api (FastAPI)	`min-inst 1, 2vCPU/1Gi, 300s`
worker	`Cloud Run Job, scale to 0`
litellm	`scale to 0`
mcp	`internal ingress only`

min-instances=1 on the API only. Cold start hits the exact request a reviewer makes, and the cross-encoder must stay resident. Everything else scales to zero. A deliberate few-dollars-for-latency trade.

MANAGED DEPENDENCIES

- **Supabase** Postgres + pgvector + auth + storage
- **Upstash** Redis — cache, queue, token buckets
- **Cloudflare R2** media, signed URLs, 5-min TTL
- **Model Armor** behind a shield adapter, 250ms budget

OBSERVABILITY

OpenTelemetry → Langfuse + Phoenix. OTEL-first so no vendor is load-bearing. One trace per ticket, parallel council members on a shared track.

ALERTS THAT MATTER — THE REST IS NOISE

- P0 recall drop on the shadow set
- Groundedness 24h rolling below band
- Escalation rate ±50% WoW — **a drop means overconfidence**
- Any fair-housing block reaching a human

CI/CD

`lint → types → unit → integration →`